Mean-logistic large-step bounds: an exact counterexample and valid repairs

Abstract

We give an admissible zero-start counterexample to the literal mean-logistic averaged-iterate bound in a pinned published theorem: the actual risk exceeds 397 where the claimed upper bound is below one. We isolate the inverse-mean convexity error, establish a parameterized sample-size delay, prove the valid exponential analogue, and repair asymptotic convergence of the unchanged logistic algorithm using a convex sum transform. Finite guarantees are proved for explicitly specified best and weighted outputs, with the remaining arithmetic-average and last-iterate limitations made explicit.

Keywords: logistic loss; large-step gradient descent; averaged iterates; counterexample; weighted convergence bounds; convex sum transform

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1 Pinned source and literal algorithms

We distinguish a false finite-time conclusion from a valid asymptotic convergence theorem. The source is Zhang, Wu, Lin and Bartlett [1], JMLR 27(124):1–31 (2026), specifically the published PDF with SHA-256

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`45880ceaef0dd8b7188f5c9c12289`.

Its masthead records submitted August 2025, revised March 2026, published April 2026. All attribution below concerns those exact bytes, without assuming that a later revision is identical. Equation (1) and Assumption 1 are on printed page 2; the adaptive rules (2)–(3) on page 5; Theorem 2 on page 6; its proof on page 9; and Lemma 16 on page 17.

Write $a_i = y_i x_i \in \mathbb{R}^d$, $\|a_i\| \le 1$. Assume a unit separator u with $a_i^T u \ge \gamma > 0$. Unless otherwise stated, γ denotes the maximum margin: if $K = \operatorname{conv}\{a_i\}$ and $a_* = \operatorname{argmin}_{a\in K}\|a\|$, then $\gamma = \|a_*\| > 0$ and $u = a_*/\gamma$. Projection gives $a_i^T a_* \ge \gamma^2$, and a convex representation of a_* shows $\min_i a_i^T u = \gamma$.

The logistic risk and exact iteration are

$$L(w) = \frac{1}{n}\sum_i \log(1 + e^{-a_i^T w}), \qquad w_{t+1} = w_t - \frac{\eta}{1 - e^{-L(w_t)}}\nabla L(w_t), \quad w_0 = 0, \tag{1}$$

where $\eta > 0$ is arbitrary. The source Theorem 2 asserts, for every integer $t \ge 1$,

$$L(\bar{w}_t) \le \exp\left(-\eta\frac{[\gamma^2(t+1)]^2 - 1}{4\gamma^2(t+1)}\right), \qquad \bar{w}_t = \frac{1}{t+1}\sum_{k=0}^{t} w_k. \tag{2}$$

There is no balance, distinct-sample, small-risk, sample-size or upper-stepsize restriction in that theorem. The averaging includes the zero iterate. We will disprove (2) for its literal mean-logistic branch. The exponential branch, with $L_{\exp} = n^{-1}\sum_i e^{-a_i^T w}$ and update $w^+ = w - \eta\nabla L_{\exp}/L_{\exp}$, is treated separately.

2 Why the inverse-mean convexity argument fails

For logistic loss $\ell(z) = \log(1 + e^{-z})$, the source uses global convexity of $\phi(w) = \log(e^{L(w)} - 1)$. The cited Ji–Telgarsky result [2] concerns an inverse of a *sum*, not an inverse of the mean. In the decreasing-margin convention its relevant transform is $\log(e^{\sum_i \ell(z_i)} - 1)$. In their appendix the superadditivity step has the form $F(\sum s_i) \geq \sum F(s_i)$, where for logistic loss $F(s) = e^s - 1$; it does not imply $F(\operatorname{mean} s_i) \geq \operatorname{mean} F(s_i)$. In fact Jensen gives the opposite inequality.

Proposition 2.1. The claimed inverse-mean transform is not globally convex, even on normalized strictly separable data.

Proof. Take $n = 2$, margins $(z_1, z_2) = (-\log 16, \log 16)$, and $L = \log(17/4)$. For $h(L) = \log(e^L - 1)$,

$$h' = 17/13, \quad h'' = -68/169, \quad \partial_1 L = -8/17, \quad \partial_{11} L = 8/289.$$

Thus $\partial_{11}(h \circ L) = 104/2873 - 256/2873 = -152/2873 < 0$. The loss hypotheses hold: $\ell'^2/(\ell\ell'') = e^{-z}/\log(1 + e^{-z})$ is decreasing, since $r/\log(1 + r)$ increases for $r > 0$. Realize these margins with $a_1 = e_1$, $a_2 = e_2$, a normalized separable dataset with maximum margin $1/\sqrt{2}$. The witness need not lie on its zero-start orbit; by itself it is a proof-lemma counterexample, not yet an algorithmic counterexample. ∎

3 An actual averaged-iterate counterexample

Theorem 1. *For $n = 1000$, $\eta = 10^6$, 999 copies of $a = (1, 0)$ and one copy of $b = (-1/2, \sqrt{3}/2)$, all with label $+1$, the exact zero-start recurrence (1) satisfies $L(\bar{w}_{200}) > 397$. Its maximum margin is $1/2$. Thus the published mean-logistic conclusion (2) is false.*

Proof. Both vectors are unit and $u = a + b$ is unit with both margins $1/2$. For any unit v, $\min(a^T v, b^T v) \leq (a + b)^T v/2 \leq 1/2$, establishing the exact margin. Write $p = 999/1000$, $q = 1/1000$, $A(w) = a^T w$, $B(w) = b^T w$, $s(z) = (1 + e^z)^{-1}$ and $D(w) = 1 - e^{-L(w)}$. The direction is

$$w^+ - w = \eta\{ps(A(w))a + qs(B(w))b\}/D(w).$$

At zero, $L = \log 2$, $D = 1/2$ and $s = 1/2$, so $w_1 = \eta(pa + qb)$. Define $v_0 = 0$ and $v_k = w_1 + (k - 1)1000b$ for $1 \leq k \leq 200$. Exactly

$$A(v_k) = 998500 - 500(k - 1) \geq 899000, \quad B(v_k) = -498500 + 1000(k - 1) \leq -299500.$$

We prove $\|w_k - v_k\| \le (k-1)/1000$. The base is exact. If the estimate holds at $k \le 199$, its error is below one, hence $A(w_k) > 898999$, $B(w_k) < -299499$ and $L(w_k) > 299$. Thus $e^{-L} < e^{-299} < 1/2$, $1/D < 2$, $s(A) < e^{-299}$ and $1 - s(B) < e^{-299}$. Subtracting the ideal increment gives

$$\|w_{k+1} - w_k - 1000b\| < 2\eta(p+2q)e^{-299} < 4{,}000{,}000e^{-299} < 1/1000.$$

The last inequality follows from $e^{299} > 2^{32} > 4{,}000{,}000{,}000$. This closes the induction without a floating-point orbit or assumed sign itinerary.

The exact rare-margin sum is $\sum_{k=1}^{200} B(v_k) = -79{,}800{,}000$, while $\|\bar{w}_{200} - \bar{v}_{200}\| \le 19900/201000 < 1$. Therefore

$$L(\bar{w}_{200}) \ge -qB(\bar{w}_{200}) > \frac{79{,}800{,}000/201 - 1}{1000} = \frac{79{,}799{,}799}{201{,}000} > 397.$$

At these same parameters the source bound is

$$\exp(-10^6 \cdot 40385/3216) = \exp(-2524062500/201) < 1,$$

a contradiction. Every finite iterate is well defined because finite w has finite positive loss and $0 < D(w) < 1$. All source hypotheses, including initialization and averaging, have been matched. ∎

This is stronger than identifying a gap in a proof: the declared averaged output violates the declared bound. It makes no claim against the exponential branch or any later corrected statement.

4 A parameterized sample-size delay

Theorem 2. *Let $n \ge 8$, take $n-1$ copies of a and one of b above, and let $\eta \ge 8n\log(32n)$. Put $T = \lfloor n/4 \rfloor$. For $1 \le k \le T$, the actual orbit has $B(w_k) \le -\eta/8$ and $L(w_k) \ge \eta/(8n)$. For $1 \le t \le T$,*

$$L(\bar{w}_t) \ge \frac{\eta}{8n}\frac{t}{t+1} \ge \frac{\eta}{16n} > 1.$$

The best loss among $w_0, \dots, w_T$ is exactly $\log 2$. The maximum margin remains $1/2$.

Proof. Put $p = 1 - 1/n$, $q = 1/n$, $v_k = \eta(pa + kqb)$ and $E = e^{-\eta/(8n)} \le 1/(32n)$. The ideal margins are

$$A(v_k) = \eta[1 - (k+2)/(2n)] \ge 3\eta/4, \quad B(v_k) = \eta[-1/2 + (k+1/2)/n] \le -3\eta/16.$$

Inductively $\|w_k - v_k\| \le 4\eta(k-1)E \le \eta(k-1)/(8n) \le \eta/32$. Indeed this error gives actual bounds $A \ge 23\eta/32$, $B \le -5\eta/32$, $L \ge \eta/(8n)$; hence $e^{-L} \le E$, $1/D \le 2$, $s(A) \le E$, $1 - s(B) \le E$. The direction differs from qb by at most $4E$. This proves the induction. Its

weaker bound $B \le -\eta/8$ yields the last-iterate risk bound, and linearity of B in the zero-inclusive average gives the asserted average bound directly. Since $\eta/(16n) \ge \frac{1}{2}\log(32n) > 1 > \log 2$, the best-iterate assertion follows. ∎

For any prescribed integer $B_0 \ge 1$, choose $n \ge \max(8, 4B_0)$. The delay lasts at least B_0 for every η on the stated half-line. Thus no sample-size-independent, margin-only burn-in can hold uniformly for all sufficiently large base stepsizes. For $n \ge 16$ and $4 \le t \le T$, the coefficient in (2) is $((t+1)^2-16)/(16(t+1)) > 0$, providing a whole family of source-theorem contradictions. The sufficient threshold is not claimed optimal.

5 The valid arbitrary-base exponential theorem

Theorem 3. *For normalized separable data, $w_0 = 0$, and any $\eta > 0$, normalized exponential descent satisfies, with $A_{\exp} = 2\eta \log n + \eta^2(1+\gamma)^2$,*

$$\|w_t - \eta t a_*\|^2 \le A_{\exp} t, \qquad L_{\exp}(w_t) \le e^{-\eta\gamma^2 t + \sqrt{A_{\exp} t}}.$$

Moreover $w_t/t \to \eta a_*$, $w_t/\|w_t\| \to u$, and $-\log L_{\exp}(w_t)/t \to \eta\gamma^2$. For $t \ge 4A_{\exp}/(\eta^2\gamma^4)$ the loss is at most $e^{-\eta\gamma^2 t/2}$.

Proof. The direction is $B = \sum_i p_i a_i$, where $p_i = e^{-a_i^T w}/\sum_j e^{-a_j^T w}$ are probabilities. Entropy gives

$$\langle w, B\rangle = H(p) - \log \sum_i e^{-a_i^T w} \le \log n + \min_i a_i^T w \le \log n + a_*^T w.$$

Projection gives $a_*^T B \ge \gamma^2$. For $e_t = w_t - \eta t a_*$, therefore $\langle e_t, B_t - a_*\rangle \le \log n$ and $\|B_t - a_*\| \le 1 + \gamma$. Expanding $e_{t+1} = e_t + \eta(B_t - a_*)$ proves the tracking bound by induction. Every margin is at least $\eta\gamma^2 t - \sqrt{A_{\exp} t}$, proving the risk upper bound and velocity limit. Some index j satisfies $a_j^T a_* = \gamma^2$, because a convex representation of a_* averages projection inequalities to equality. Consequently

$$L_{\exp}(w_t) \ge n^{-1} e^{-\eta\gamma^2 t - \sqrt{A_{\exp} t}},$$

which proves the exact risk exponent. The burn-in follows by solving $\sqrt{A_{\exp} t} \le \eta\gamma^2 t/2$. ∎

For logistic descent the analogous weights are $c_i = [n(1-e^{-L})(1+e^{a_i^T w})]^{-1}$ and need not sum to one. Replacing them by probabilities would change the algorithm and erase the rare-sample delay above.

6 A valid repair for the actual logistic dynamics

Define the sum transform, cumulative effective step, and constants

$$\Psi(w) = \log(e^{nL(w)} - 1), \quad \lambda_t = \frac{\eta}{n}\sum_{j=0}^{n-1} e^{-jL(w_t)}, \quad S_t = \sum_{j<t} \lambda_j,$$
$$C_n = \log(2^n - 1), \quad A = 2\eta C_n + \eta^2(1+\gamma)^2, \quad b_0 = \eta\gamma^2/n.$$

Theorem 4. *The literal recurrence* (1) *exists for every finite time and every* $\eta > 0$. *It satisfies*

$$\eta/n \le \lambda_t \le \eta, \quad \|w_t - \gamma S_t u\|^2 \le At,$$
$$L(w_t) \le e^{-\gamma^2 S_t + \sqrt{At}} \le e^{-b_0 t + \sqrt{At}}.$$

In particular $L(w_t) \le e^{-b_0 t/2}$ *for* $t \ge 4A/b_0^2$. *Furthermore*

$$w_t/t \to \eta\gamma u, \quad w_t/\|w_t\| \to u, \quad -\log L(w_t)/t \to \eta\gamma^2.$$

The finite deficit $D = \sum_{t\ge 0}(\eta - \lambda_t)$ *exists, and*

$$\|w_t - \eta\gamma t u\| \le \sqrt{At} + \gamma D, \quad L(w_t) \le e^{-\eta\gamma^2 t + \sqrt{At} + \gamma^2 D}.$$

Proof. For nonempty subsets $J \subseteq \{1, \dots, n\}$ put $a_J = \sum_{i\in J} a_i$. The exact product expansion gives

$$e^{nL(w)} - 1 = \prod_i (1 + e^{-a_i^T w}) - 1 = \sum_{J\ne\varnothing} e^{-a_J^T w}.$$

Thus Ψ is a finite log-sum-exp of affine functions, hence convex. Singletons and the maximum-margin geometry give $\Psi(w) \ge -\gamma u^T w$. For $r \ge 0$, every subset has $a_J^T r u \ge \gamma r$, so $\Psi(ru) \le -\gamma r + C_n$.

Differentiation gives $\nabla\Psi = n\nabla L/(1 - e^{-nL})$. Thus the exact update is $w_{t+1} = w_t - \lambda_t \nabla\Psi(w_t)$, with the stated geometric-sum expression and bounds. Let $\ell_i = \log(1 + e^{-a_i^T w})$ and $h(x) = 1 - e^{-x}$. Concavity of h gives

$$\|\nabla L\| \le n^{-1} \sum_i h(\ell_i) \le h(L),$$

so every actual step has norm at most η. This proves finite existence and excludes zero denominators. Also its projection on u is strictly positive, so $w_t \ne 0$ for $t \ge 1$.

Set $e_t = w_t - \gamma S_t u$, $g_t = \nabla\Psi(w_t)$ and $z_t = \gamma S_t u$. Convexity and the two affine estimates give

$$\langle g_t + \gamma u, e_t\rangle \ge \Psi(w_t) - \Psi(z_t) + \gamma(u^T w_t - \gamma S_t) \ge -C_n.$$

Also $\|\lambda_t(g_t + \gamma u)\| \le \eta(1 + \gamma)$. Expanding the exact error recurrence proves $\|e_{t+1}\|^2 \le \|e_t\|^2 + A$. Every margin is at least $\gamma^2 S_t - \sqrt{At}$; using $\log(1 + x) \le x$ proves both risk bounds and the explicit burn-in.

Thus $L(w_t) \to 0$, $\lambda_t \to \eta$, and by Cesaro $S_t/t \to \eta$. The tracking bound proves the velocity and direction limits. If $m_t = \min_i a_i^T w_t$, then $m_t/t \to \eta\gamma^2$, so eventually $m_t \ge 0$. The inequalities $e^{-m_t}/(2n) \le L(w_t) \le e^{-m_t}$ prove the exact exponent. Finally

$$0 \le \eta - \lambda_t \le \eta(n - 1)L(w_t)/2.$$

The exponential tail makes $\sum L(w_t)$ finite, hence $D < \infty$. Since $0 \le \eta t - S_t \le D$, the remaining bounds follow. ∎

This proof establishes eventual convergence of the actual algorithm despite the false source finite-time estimate. Its coarse sample-size-dependent bound is not a solution of the sharp arithmetic-average or last-iterate upper-bound question.

7 Best-iterate and effective-step-weighted finite bounds

For $T \ge 1$ define $w_T^\lambda = S_T^{-1} \sum_{t<T} \lambda_t w_t$ and choose w_T^{best} among $w_0, \dots, w_{T-1}$ minimizing L. The index convention differs from the source's zero-through-t arithmetic average.

Theorem 5. *For either output $q = w_T^\lambda$ or w_T^{best} and every $r \ge 0$,*

$$\Psi(q) \le \log((1 + e^{-\gamma r})^n - 1) + \frac{(r + \eta/(2\gamma))^2}{2S_T}.$$

If $\gamma^2 S_T \ge \log n + \eta/2$, put $B_T = \log n + 1 + \eta/2 - \gamma^2 S_T/2$. Then

$$L(q) \le n^{-1} \log(1 + e^{B_T}), \qquad \min_i a_i^T q \ge -B_T.$$

For any $\rho \ge 0$, the sufficient condition

$$T \ge \frac{n}{\eta\gamma^2}(\eta + 2\log n + 2 + 2\rho)$$

gives $\min_i a_i^T q \ge \rho$ and $L(q) \le n^{-1} \log(1 + e^{-\rho})$.

Proof. Let $c_i = h(\ell_i)/(nh(L))$, $s = \sum_i c_i \le 1$, and $B = \sum_i c_i a_i$. The actual step $d = \eta B$ obeys $\|B\|^2 \le s$ and $u^T B \ge \gamma s$, hence $\|d\|^2 \le (\eta/\gamma) u^T d$. For $h_0 = \eta u/(2\gamma)$, expansion of distance to any fixed $v + h_0$ and convexity of Ψ give

$$\|w_{t+1} - v - h_0\|^2 \le \|w_t - v - h_0\|^2 - 2\lambda_t(\Psi(w_t) - \Psi(v)).$$

Summing and dropping the final squared distance yields

$$2 \sum_{t<T} \lambda_t(\Psi(w_t) - \Psi(v)) \le \|v + h_0\|^2.$$

Weighted Jensen applies to w_T^λ; the minimum is no larger than the weighted mean and Ψ increases strictly with L. Set $v = ru$ and use the product representation to obtain the first claim.

The inequality $(1+x)^n - 1 \le nx(1+x)^{n-1}$ follows by integrating $n(1+y)^{n-1}$ over $[0, x]$. Hence $\Psi(ru) \le \log n - \gamma r + ne^{-\gamma r} \le \log n + 1 - \gamma r$ if $\gamma r \ge \log n$. Choose $r = \gamma S_T - \eta/(2\gamma)$ to obtain B_T. The identities $nL = \log(1 + e^\Psi)$ and $\Psi \ge -a_i^T w$ give the risk and margin bounds. Finally use $S_T \ge \eta T/n$ to obtain the sufficient iteration count. ∎

8 Separation is not absorbing; a route limitation

Proposition 8.1. For three copies of a and two copies of b, and every $\eta \geq 10 \log 10$, the first logistic iterate is strictly separating and strictly improves the initial risk, but $a^T w_2 \leq -\eta/20$ and $L(w_2) \geq 3\eta/100$.

Proof. Exactly $w_1 = \eta(3a + 2b)/5$, whose margins are $2\eta/5$ and $\eta/10$. Set $t = e^{-\eta/10} \leq 1/10$, $p_A = t^4/(1+t^4)$, $p_B = t/(1+t)$, $L_1 = (3/5)\log(1+t^4)+(2/5)\log(1+t)$, $D = 1-e^{-L_1}$. Then

$$a^T w_2/\eta = 2/5 + (3p_A - p_B)/(5D).$$

Since $D \leq (3/5)t^4 + (2/5)t$, $p_A \leq t^4$, $p_B \geq 10t/11$,

$$p_B - 3p_A - (9/4)D \geq t/110 - (87/20)t^4 \geq t(1/110 - 87/20000) > 0.$$

Thus $a^T w_2/\eta < 2/5 - 9/20 = -1/20$. Retaining the three a samples gives the risk bound. Step one has all positive margins, so each loss is less than $\log 2$. ∎

For completeness, a saturated finite-itinerary approximation and its ceiling explain what that mechanism can and cannot prove. Define

$$v_1 = n^{-1} \sum_i a_i, \quad N_k = \{i : a_i^T v_k < 0\}, \quad b_k = n^{-1} \sum_{i \in N_k} a_i, \quad v_{k+1} = v_k + b_k.$$

Suppose $N_k \neq \varnothing$ for $1 \leq k \leq H$ and all itinerary margins through $H+1$ have magnitude at least $\delta > 0$. If $\eta \geq (2n/\delta)\log(8H/\delta)$, then, for $1 \leq k \leq H+1$,

$$\|w_k/\eta - v_k\| \leq 4(k-1)e^{-\eta\delta/(2n)} \leq \delta/2.$$

Indeed the base is exact. Under the inductive error bound every actual margin has the correct sign and magnitude at least $\eta\delta/2$. Put $E = e^{-\eta\delta/(2n)}$. Each sigmoid differs from its negative-margin indicator by at most E, and the nonempty negative set gives $L \geq \eta\delta/(2n)$, hence $e^{-L} \leq E \leq 1/8$. The actual normalized direction differs from b_k by at most $2E/(1-E) \leq 4E$, closing induction. Endpoint strict separation, if present, transfers, but is not absorbing by Proposition 8.1.

For this batch route alone, put $m_k = |N_k|$ and $S = 1 + \sum_{k=1}^{H} m_k/n$. Projection gives $u^T v_{H+1} \geq \gamma S$. Since $v_k^T b_k \leq 0$ and $\|b_k\|^2 \leq m_k/n$, expansion gives $\|v_{H+1}\|^2 \leq S$. Thus $S \leq \gamma^{-2}$ and $H \leq n(\gamma^{-2}-1)$. Some state of index at most $\lfloor n(\gamma^{-2}-1)\rfloor+1$ therefore has no negative margin; without a no-zero-margin hypothesis it need not strictly separate. This ceiling limits a particular lower-bound construction. It is not a universal actual-logistic burn-in theorem.

9 Conclusion and reproducibility

The mean-logistic source bound is contradicted at its actual declared average. The inverse-mean convexity lemma is independently false, and the sum-versus-mean transfer

is invalid. These negative conclusions coexist with rigorous eventual convergence and an exact asymptotic risk exponent for the unchanged algorithm. Valid finite bounds have been supplied for the best iterate and effective-step-weighted average; they must not be relabeled as an arithmetic-average or sharp last-iterate bound. Those sharp upper questions remain unresolved here.

The supplied exact-arithmetic script checks the Hessian witness, source-bound coefficient, averaged-risk rational contradiction, and sample-delay constants. Infinite-time claims rely on the proofs above, not on floating-point simulation. The Ji–Telgarsky published ALT PDF used for the sum/mean comparison has SHA-256 `b28cea822ec2898201ccae41778dc7315d46ac88e1bd77a2b4b401c3aabad0a7`; the corresponding arXiv v3 is dated 12 November 2020. The relevant results are published Lemma 12 and equation (26), corresponding to arXiv Lemma 5.2 and equation (E.3).

References

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[2] Z. Ji and M. Telgarsky. A refined primal-dual analysis of the implicit bias. *Proceedings of Algorithmic Learning Theory*, PMLR 132, 2021. `https://proceedings.mlr.press/v132/ji21a.html`.